

Day 3 — Transformers Interview Revision

Multi-Head Attention

High-Yield Concepts

- Why multiple attention heads are used
- Each head has its own learned Q, K, and V projections
- Each head independently performs scaled dot-product attention
- Attention outputs are concatenated along the feature dimension
- Final learned linear projection
- Returning the representation to the model / embedding dimension

Interview Questions + Answers

1. Why do we use multiple attention heads instead of one?

Multiple attention heads allow the model to learn different relationships and representations from the same sequence simultaneously. Each head has its own learned Q, K, and V projections, allowing different attention patterns to emerge.

2. What does each attention head do?

Each head has its own learned Q, K, and V projection matrices. It generates Q, K, and V from the input, calculates attention scores, applies softmax to obtain attention weights, and takes a weighted sum of the Value vectors.

3. Why not just use one large attention head?

Multiple heads allow the model to operate in different learned representation subspaces and capture different relationships simultaneously. Parallel computation is a benefit, but it is not the main reason for using multiple heads.

4. How are the outputs of multiple heads combined?

The outputs of all heads are concatenated along the feature dimension. The concatenated representation is then passed through a learned linear projection to produce the final Multi-Head Attention output.

5. Explain Multi-Head Attention from input to output.

The input X is passed into multiple attention heads. Each head has its own learned Q, K, and V projection matrices and independently computes scaled dot-product attention. The resulting head outputs are concatenated along the feature dimension and passed through a final learned linear projection to return the representation to the model dimension.

Inside One Attention Head

$$Q_h = XWQ_h$$

$$K_h = XWK_h$$

$$V_h = XWV_h$$

$$O_h = \text{softmax}(Q_h K_h^T / \sqrt{d_k}) V_h$$

Each head has its own learned WQ , WK , and WV matrices. Therefore, different heads can learn different attention patterns.

Dimension Walkthrough

Let N be the number of tokens, D the model / embedding dimension, H the number of heads, and P the dimension of each head.

Step	Dimensions	Explanation
Input X	$N \times D$	Input token representations
One head output	$N \times P$	Output produced by one attention head
H head outputs	$H \times (N \times P)$	Each head produces its own output
Concatenation	$N \times (HP)$	Concatenate along the feature dimension
Output projection	$(N \times HP)(HP \times D)$	Learned linear transformation
Final output	$N \times D$	Returns to model / embedding dimension

Simple Example

Suppose the model dimension is $D = 512$ and there are $H = 8$ heads.

- Each head has $P = 512 / 8 = 64$ features.
- Each head produces an $N \times 64$ output.
- Eight heads are concatenated $\rightarrow N \times 512$.
- The final output projection maps $N \times 512 \rightarrow N \times 512$.

Important Distinction

Within a head: attention weights are used to calculate a weighted sum of the Value vectors.

Across heads: the head outputs are concatenated. They are NOT combined using another attention-weighted sum.

Interview-Ready Summary

Multi-Head Attention runs several attention mechanisms in parallel. Each head has its own learned Q , K , and V projections and can learn different relationships in the sequence. The outputs of the heads are concatenated along the feature dimension and passed through a learned output projection to return the representation to the original model dimension.